Heat Transfer Problem Set #2

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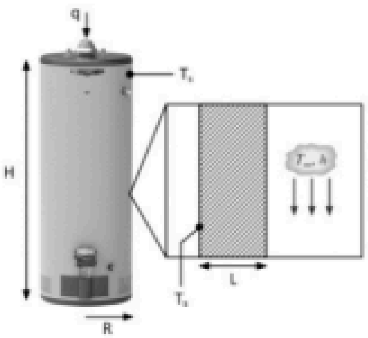
<u>Question 1</u>

1. (40 pt) Consider an electric water heater operated under a steady state. This water heater is a cylinder with radius $R = 0.5$ m and height $H = 1$ m. When the water heater is fully filled, the water temperature inside can be maintained at 50 °C. The wall of the water heater is thin enough and of high thermal conductivity, so we can assume that the exterior wall temperature of the water heater T_s is also 50 °C at the steady state. The water heater is placed in an environment with ambient temperature of $T_\infty = 20$ °C and heat transfer coefficient of $h = 15$ W/(m²·K). The bottom of the water heater is well thermally insulated. Thermal radiation is negligible in this problem.



(a) Calculate the total electrical heating power q needed to maintain a constant water temperature at 50 °C.

(b) To reduce the heat loss from the water heater, we now attach a $L = 2$ cm thick thermal insulation foam to the exterior wall of the water heater, including both side wall and top wall (zoom-in illustration). Thermal conductivity k of the insulation foam is 0.1 W/(m·K). Since the thickness of the insulation foam is much smaller than the radius of the water heater, we can assume 1D heat conduction across the insulation foam. To maintain a constant water temperature at 50 °C, now how much total electrical heating power q is needed?

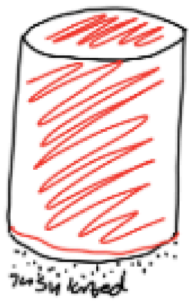
(c) Compare the results obtained from (a) and (b), discuss the role of thermal insulation in the energy saving of water heater.

Known: Heater dimensions, $T_s = 50°C$, $T_\infty = 20°C$, $h = 15\ W/(m^2 \cdot K)$

Assumptions: Insulated bottom, negligible radiation, constant air properties, steady state

Find: Heating power q to maintain temp w/ and w/o insulation.

Analysis a)



$A_{side} = 2\pi RH = 2\pi(0.5)(1) = \pi = 3.14\ m^2$

$A_{top} = \pi R^2 = \pi(0.5)^2 = 0.785\ m^2$

$A_{total} = 3.14 + 0.785 = 3.925\ m^2$

$q = hA(T_s - T_\infty) = (15\ W/(m^2 \cdot K))(3.925 m^2)(50-20)$

$\boxed{= 1766.25\ W}$

Analysis b) $L = 2cm$, $k = 0.1\ W/(m \cdot K)$, assume 1D heat conduction

transfer occurs by conduction thru insulation and convection from insulation outer surface

By thermal resistances:

$$R_{cond} = \frac{L}{kA} = \frac{0.02}{(0.1)(3.925)} = 0.051\ K/W$$

$$R_{conv} = \frac{1}{hA} = \frac{1}{(15)(3.925)} = 0.017\ K/W$$

$$R_{total} = 0.051 + 0.017 = 0.068\ K/W$$

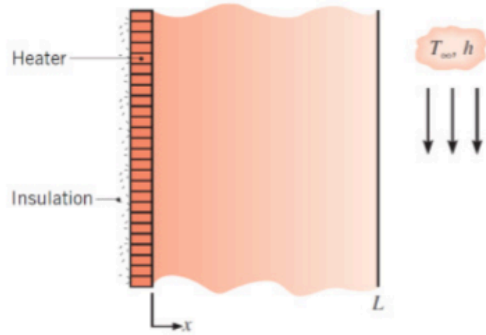
$$q = \frac{T_w - T_\infty}{R_{total}} = \frac{50-20}{0.068} = \boxed{441.18 W}$$

Analysis c) $1 - \frac{441.18}{1,766.25} \approx 75\%$

Thermal insulation significantly increases the thermal resistance between the water and environment. It results in a ~75% reduction in heat loss, indicating that the energy efficiency is much improved with the addition of insulation.

Question 2

2. (40 pt) A thin film heater is sandwiched by a thermally insulating wall and a L = 10 cm thick plastic layer with a thermal conductivity of k = 1 W/(m·K). The exterior surface of the plastic layer is immersed in a coolant with temperature T_∞ = 20 °C and heat transfer coefficient h = 100 W/(m²·K). The system is at a steady state. A 1D coodinate system is shown in the right figure. Assume the heater is infinitely thin and neglect thermal radiation.



(a) When the thin film heater supplies a heat flux of q'' = 1000 W/m², calculate the temperature of the interior surface (x = 0) and the exterior surface (x = L) of the plastic layer.

(b) To avoid overheating, the temperature of the heater cannot exceed 200 °C. Calculate the maximum heat flux that can be provided by the thin film heater without the concern of overheating.

(c) Discuss three strategies that enable the thin film heater to be operated under even higher heat flux without exceeding 200 °C.

Known: $L = 10\,cm$, $k = 1\,W/(m\cdot K)$, $T_\infty = 20°C$, $h = 100\,W/(m^2\cdot K)$,
$q'' = 1000\,W/m^2$ (part a), $T_{max} = 200°C$ (part b)

Assumptions: 1D conduction, constant properties, steady state conditions, uniform heat flux.

Find: Interior/Exterior temps, max heat flux

Analysis a) By thermal resistances:

$$R_{cond} = \frac{0.1}{1} = 0.1\ m^2K/W$$

$$R_{conv} = \frac{1}{100} = 0.01\ m^2K/W$$

$$R_{total} = 0.10 + 0.01 = 0.11\ m^2K/W$$

@ $x = L$ $\quad q'' = h(T_L - T_\infty)$

$$1000 = 100(T_L - 20) \rightarrow \boxed{T_L = 30°C}$$

@ $x = 0$ $\quad \Delta T = q''\frac{L}{k} = 1000(0.10) = 100°C$

$$T_0 = T_L + 100 \rightarrow 30 + 100 = \boxed{130°C}$$

Analysis b) $T_0 = T_\infty + q'' R_{total}$

$$200 = 20 + q''(0.11)$$

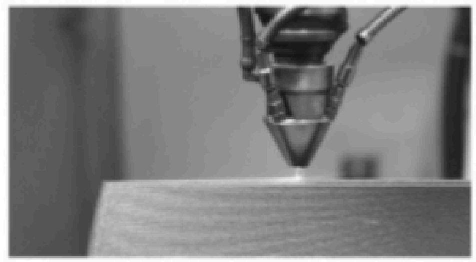
$$180 = 0.11 q'' \rightarrow \boxed{q''_{max} = 1636.36\ W/m^2}$$

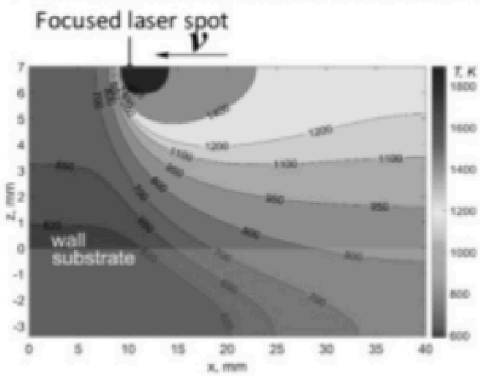
Analysis c)

The first strategy could be to reduce plastic thk. L. This would lower the conduction resistance and temp increases. Another strategy could be to use higher conductivity (k) material. This would reduce conduction resistance to allow for higher heat flux. A final strategy could be to increase coefficient h to have stronger inherent cooling and lower R_{conv}. In turn this would also allow an increase in heat flux w/o exceeding 200°C.

Question 3

3. (20 pt) Laser based additive manufacturing is an emerging technique to create complex 3D metal structures with high spatial resolution. During the manufacturing process, a high-power laser beam is focused on a pool of metal powders. By scanning the focused laser beam, metal powders under the irradiation can absorb the laser energy, melt, and re-solidify, thereby creating dense solid metal with desirable structures. The right figure shows a representative cross-section temperature profile (isotherm) of a metal substrate under focused laser heating.





(a) Consider 2D heat transfer within the x-z plane, please schematically draw the heat flux vectors based on the given temperature profile in the right figure.

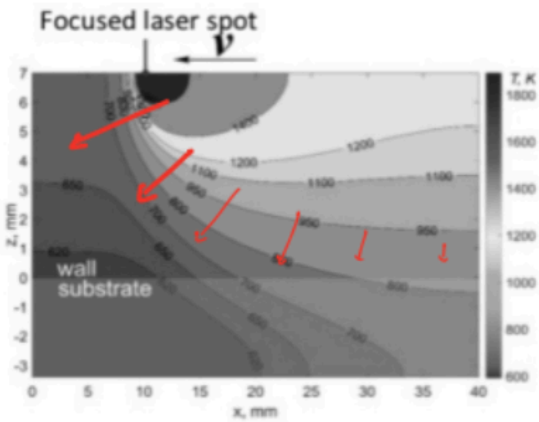
(b) Based on the temperature profile, please briefly explain which side the laser spot (left or right) has larger heat flux.

Known: 2D heat transfer in X-Z plane, isotherms given

Assumptions: Steady state conduction, constant thermal conductivity, heat flux vectors are perpendicular to isotherms.

Find: schematic heat flux vectors, higher heat flux side.

Analysis a)



Heat flux vectors are perpendicular to the temperature contours and heat is flowing from warmer regions to cooler regions. The vectors have greater magnitude where the contours are closer together (more abrupt temp changes). Steeper gradient corresponds w/ aggressive flux vectors.

Analysis b) $|\vec{q}| \propto |\nabla T|$

The left side of the laser spot has larger heat flux because the temperature contours are much closer together on that side, indicating a larger gradient and more heat flux. The right side shows gradients that are spaced more widely.